

TECHNICAL RESEARCH REPORT

RF-RSSI Based Turbidity Sensing

Hypothesis, Theoretical Basis, Experimental Results,
Findings and Alternative Strategies

Solid-State Water Quality Testing Cup Project

VESIT, Mumbai University | Electronics and Telecommunication Engineering
B.Tech Batch 2024-2028

Table of Contents

- 1. Hypothesis 3
- 2. Theoretical Basis 3
 - 2.1 The Physics of 2.4 GHz RF in Water 3
 - 2.2 Expected Effect of Turbidity on RSSI 4
 - 2.3 Why NTU Cannot Be Derived Theoretically..... 4
 - 2.4 The Empirical Calibration Approach 4
- 3. Experimental Setup 5
 - 3.1 Components Used 5
 - 3.2 Pin Connections and Wiring 5
 - 3.3 Firmware Architecture 6
 - 3.4 Procedure..... 7
- 4. Experimental Results..... 7
- 5. Analysis of Failure Modes 8
 - 5.1 Signal Direction Problem..... 8
 - 5.2 Signal Magnitude Problem 9
 - 5.3 Multipath Interference 9
 - 5.4 Physical Setup Deficiencies 9
- 6. Alternative Strategy A: RF Shielding (Aluminium Foil)..... 10
- 7. Alternative Strategy B: CSI Phase Measurement 11
- 8. Comparative Assessment 13
- 9. Conclusion and Recommended Path Forward 13

1. Hypothesis

This investigation was founded on the following hypothesis:

Primary Hypothesis

The received signal strength (RSSI) of a 2.4 GHz ESP-NOW radio link transmitted through a water column will decrease monotonically and measurably as the turbidity of that water column increases, such that RSSI may be used as a surrogate measurement for turbidity in NTU after empirical calibration.

This hypothesis was motivated by the broader design mandate of the Solid-State Water Quality Testing Cup project, which requires all sensing subsystems to operate without any electrical contact between sensing electronics and the water under test. Optical turbidity sensors require submerged optical components that are susceptible to fouling. The RF method, by placing radio transceivers on the exterior of the cup wall, offers a completely non-contact architecture and was therefore selected for investigation as the turbidity sensing channel.

2. Theoretical Basis

2.1 The Physics of 2.4 GHz RF in Water

Water is a polar molecule. Its oxygen and hydrogen atoms carry partial negative and positive charges respectively, forming a permanent electric dipole. When an electromagnetic wave at 2.4 GHz passes through liquid water, it exerts an oscillating electric field that forces these dipole molecules to reorient 2.4 billion times per second. This rotational friction dissipates kinetic energy from the wave as heat, attenuating the RF signal.

This phenomenon is quantified by the complex relative permittivity of water:

$$\epsilon_r = \epsilon' - j\epsilon''$$

Where ϵ' is the real part (dielectric constant, governing wave velocity) and ϵ'' is the imaginary part (dielectric loss factor, governing attenuation). At 2.4 GHz and 25 degrees Celsius, pure water exhibits values of approximately:

Parameter	Symbol	Value at 2.4 GHz, 25 C
Dielectric constant	ϵ'	~78.4

Parameter	Symbol	Value at 2.4 GHz, 25 C
Dielectric loss factor	ϵ''	~12.3
Loss tangent	$\tan(\delta) = \epsilon''/\epsilon'$	~0.157
Attenuation coefficient	a	~295 dB/m

The practical consequence is that a 2.4 GHz radio signal traversing 5 cm of pure water experiences approximately 14.75 dB of attenuation relative to the same path through air. This provides a strong, clearly measurable baseline signal drop when the water-filled cup is placed between the two transceivers.

2.2 Expected Effect of Turbidity on RSSI

When suspended solid particles are introduced into the water column, two mechanisms were theorised to affect the RF signal. First, particle scattering was expected to deflect RF energy off-axis, reducing the power reaching the receiving antenna. Second, dielectric displacement was expected to alter the bulk permittivity of the water-particle mixture, changing the attenuation characteristics of the medium. Together, these mechanisms were expected to produce a decrease in RSSI (a more negative dBm value) as turbidity increased.

2.3 Why NTU Cannot Be Derived Theoretically

NTU (Nephelometric Turbidity Units) is a strictly optical standard. Under EPA Method 180.1, NTU is defined as the quantity of light scattered at exactly 90 degrees by a suspension when illuminated by a source of specific wavelength (typically 850 nm infrared). Because this definition is rooted in optical scattering behaviour rather than electromagnetic absorption, no direct theoretical relationship between 2.4 GHz RSSI and NTU can be derived from Maxwell's equations alone. Any conversion between the two quantities must therefore be established empirically through calibration against a reference optical nephelometer.

2.4 The Empirical Calibration Approach

The proposed calibration methodology followed the Beer-Lambert analogy for signal attenuation through a medium. Because RSSI is already expressed on a logarithmic scale (dBm) and NTU is expressed on a linear scale, a quadratic polynomial regression was expected to yield the best fit between the two quantities:

$$NTU = A * RSSI^2 + B * RSSI + C$$

Where A, B, and C are empirically determined calibration coefficients unique to each device geometry. Additionally, a temperature compensation term was incorporated to account for the well-documented temperature dependence of water's dielectric loss factor:

$$RSSI_{compensated} = RSSI_{raw} - (K_{temp} * (T_{water} - 25.0))$$

Where K_{temp} is a device-specific coefficient expressing RSSI shift per degree Celsius deviation from the 25 degrees Celsius calibration reference, and T_{water} is the measured water temperature. This compensation is applied to the raw RSSI value before the NTU polynomial is evaluated.

3. Experimental Setup

3.1 Components Used

Component	Specification	Quantity	Role
ESP32-C3 Supermini	Single-core RISC-V, 2.4 GHz WiFi, ESP-IDF v5	2	RF Master (Tx) and Slave (Rx)
PET Bottle	Standard 500 mL, 50 mm inner diameter	1	Water containment vessel
USB-C Cables	Data-capable, not charge-only	2	Power and firmware flashing
USB Power Bank	5V output	1	Independent power for Master node
Laptop (PC)	Windows, Arduino IDE 2.x	1	Serial Monitor and firmware upload
Tape	Adhesive tape	1	Securing boards to bottle surface
Flour / Clay	Common household flour	1	Turbidity surrogate for testing
Tap water	Municipal supply, Mumbai	-	Test medium

3.2 Pin Connections and Wiring

The two-board system required no inter-board wiring beyond a shared power ground. Each board operated independently via its own USB connection. The following table summarises the pin usage on each node.

MASTER NODE (ESP32-C3 SUPERMINI NO. 1 — TRANSMITTER)

Pin	Function	Connected To	Notes
USB-C	Power and flashing	USB Power Bank (after flashing)	No GPIO connections required
GPIO8 (onboard LED)	Status indicator	Onboard LED (internal)	Blinks 0.5 ms per packet sent
GPIO9 (BTN_BOOT)	Boot button	Internal pull-up	Renamed from BOOT_PIN to avoid esp32-hal.h conflict
All others	Unused	Unconnected	No peripheral wiring on this node

SLAVE NODE (ESP32-C3 SUPERMINI NO. 2 — RECEIVER AND PROCESSOR)

Pin	Function	Connected To	Notes
USB-C	Power and Serial Monitor	Laptop USB port	Stays connected to PC throughout test
GPIO8 (onboard LED)	Connection status indicator	Onboard LED (internal)	Fast blink = receiving; slow blink = offline
GPIO9 (BTN_BOOT)	Boot button	Internal pull-up	Same rename convention as Master
All others	Unused	Unconnected	No peripheral wiring on this node

Neither board required any external resistors, capacitors, or passive components for this experiment. Decoupling capacitors (100 uF electrolytic and 100 nF ceramic) were specified in the project design but were unavailable at the time of testing. This absence was partially compensated by increasing the RSSI median filter buffer size from 20 to 40 samples and by powering each board from a separate USB source to eliminate shared power rail noise coupling.

3.3 Firmware Architecture

Both nodes ran custom firmware written in Arduino-style C++ targeting the ESP32 Arduino core version 3.3.3, built on top of ESP-IDF v5. The use of core version 3.x was mandatory because the ESP-NOW receive callback signature changed between v2 and v3, and the `rx_ctrl->rssi` field is only exposed through the new `esp_now_recv_info_t` struct available in v3.

Key firmware implementation details:

- Master firmware: broadcast ESP-NOW beacon packets at 20 packets per second (50 ms interval) at maximum transmit power (21 dBm). The send callback used the updated `wifi_tx_info_t*` parameter type required by core v3.
- Slave firmware: received packets via the `esp_now_recv_cb` callback and extracted RSSI from `recv_info->rx_ctrl->rssi` as a signed 8-bit integer in dBm. Values outside the physical range of -120 to 0 dBm were rejected.
- A 40-sample circular ring buffer stored incoming RSSI values. An insertion sort median filter was applied to this buffer every 1000 ms to produce a stable filtered RSSI output.
- Temperature compensation was implemented as: $\text{RSSI_comp} = \text{RSSI_raw} - (0.2 * (T - 25.0))$, applied before the NTU polynomial evaluation.
- Both nodes locked to WiFi channel 1 via `esp_wifi_set_channel()` to prevent channel drift during measurement.

A known issue was encountered and resolved during development: the Arduino ESP32 core v3.3.x defines `BOOT_PIN` as a reserved symbol in `esp32-hal.h`. Any user code that attempts to define a variable or constant with the same name produces a redefinition compile error. This was resolved by renaming all references to `BTN_BOOT` throughout the firmware.

3.4 Procedure

The experimental procedure consisted of five sequential tests conducted in a single session:

- Test 1: Both boards placed on a desk in open air, 10 cm apart. RSSI was recorded to establish the air baseline.
- Test 2: A 500 mL PET bottle (50 mm inner diameter) filled with clean tap water was placed between the two boards, with each board's antenna face pressed against the opposite outer walls of the bottle using adhesive tape. RSSI was recorded to establish the clean-water baseline.
- Test 3: Approximately one quarter teaspoon of wheat flour was dissolved in the water. The bottle was gently agitated to produce a uniform suspension. RSSI was recorded.

- Test 4: Additional flour was added incrementally, increasing turbidity. RSSI was recorded at each increment.
- Test 5: Both hands were cupped around the bottle during measurement to attempt to block reflected and side-path RF signals. RSSI was recorded.

4. Experimental Results

Test Condition	RSSI Latest (dBm)	RSSI Median/40 (dBm)	Packets Received	Drop Rate
Open air, 10 cm separation	-29	-29	263	0.0%
Tap water, correct geometry	-38	-38	1497	9.2%
Turbid water (flour added)	-26	-25	1027	2.7%
Increased turbidity	-25	-25	1027	2.7%
Hands cupped around bottle	No significant change from Test 3			

The results immediately contradicted the primary hypothesis. Upon addition of flour to produce turbidity, the RSSI value increased (became less negative) rather than decreasing. Specifically, RSSI rose from -38 dBm in clean water to approximately -25 dBm in turbid water, representing a 13 dBm increase in received signal strength. This trend was consistent across multiple flour additions and did not reverse when the bottle was held in cupped hands to reduce reflected paths.

A secondary observation was the elevated packet drop rate of 9.2% during the clean-water test on WiFi channel 1, which reduced to 2.7% during the turbidity test. The drop rate improvement was likely attributable to the same physical mechanism driving the RSSI increase, rather than any change in link conditions.

5. Analysis of Failure Modes

5.1 Signal Direction Problem

The fundamental error in the original hypothesis was the assumption that suspended particles increase RF attenuation. At 2.4 GHz, the primary attenuator in the measurement path is water itself, not the particles. When suspended solids are introduced, they physically displace water molecules from the measurement volume. Because the solid particles (flour, clay, silt) exhibit significantly lower dielectric loss at 2.4 GHz than liquid water, replacing water volume with particle volume reduces the bulk dielectric loss of the mixture.

The effective complex permittivity of a water-particle mixture is governed by effective medium theory. For a dilute suspension, the mixture permittivity can be approximated by the Maxwell Garnett relation:

$$\epsilon_{\text{mix}} = \epsilon_{\text{water}} * [1 + 3*f*(\epsilon_p - \epsilon_{\text{water}}) / (\epsilon_p + 2*\epsilon_{\text{water}} - f*(\epsilon_p - \epsilon_{\text{water}}))]$$

Where f is the volume fraction of particles and ϵ_p is the permittivity of the particle material. Since the imaginary part of ϵ_p (the loss factor) for clay and silica is substantially lower than that of water, increasing f

decreases the imaginary part of ϵ_{mix} , which reduces attenuation. The net measured effect, as confirmed experimentally, is that RSSI increases with turbidity. This is the physically correct response.

5.2 Signal Magnitude Problem

Even if the signal direction problem were somehow corrected, a second fundamental constraint prevents the RSSI method from achieving useful precision at drinking water turbidity levels. At 100 NTU (a moderately turbid sample), the volume fraction of suspended particles is approximately:

$$f = (50 \text{ mg/L}) / (2600 \text{ mg/cm}^3) = 0.0019\%$$

This is less than 0.002 percent of the water volume. The resulting change in bulk dielectric loss is far below the 1 dBm resolution limit of the ESP32's RSSI measurement. The signal of interest is buried beneath the noise floor. Extending the path length can partially mitigate this: signal change scales linearly with path length, and at 20-30 cm the effect approaches the detectable threshold. However, a 20-30 cm path length is incompatible with the cup-form-factor design goal.

5.3 Multipath Interference

In an open bench environment, the received signal at the Slave antenna is the vector sum of multiple propagation paths: the direct path through the water column, reflected paths from the desk surface, from the laptop screen, and from surrounding walls. As turbidity changes and the direct through-water path is altered, these reflected paths may constructively or destructively interfere in unpredictable ways. In the experimental results, the RSSI increase observed with turbidity may partly reflect a change in interference conditions rather than a direct dielectric effect. Without shielding all non-direct paths, the measurement is not geometrically controlled.

5.4 Physical Setup Deficiencies

Two specific physical setup problems were identified through visual inspection of the experimental photographs:

- **Air gap from pin headers:** The ESP32-C3 Supermini boards were held against the bottle surface using header pins as spacers. This created an air gap of approximately 8-12 mm between the PCB antenna and the bottle surface. This gap allowed RF energy to leak around the bottle through air rather than passing through the water column, reducing the proportion of received signal that had actually traversed the measurement medium.
- **Insufficient RF path control:** The 50 mm diameter bottle was insufficiently wide to produce a dominant through-water signal at 2.4 GHz. The signal travelling around the bottle through air (approximately 130 mm semi-circumference, very low loss) competed directly with the signal through 50 mm of water (high loss). The net received signal was dominated by the low-loss air path rather than the through-water path of interest.

6. Alternative Strategy A: RF Shielding with Aluminium Foil

The first alternative strategy addresses the multipath interference and air-bypass problems without changing any hardware or firmware. It requires only a sheet of aluminium foil.

6.1 Principle of Operation

Aluminium foil acts as a Faraday shield for 2.4 GHz RF signals. Its conductivity (approximately 3.8×10^7 S/m) is high enough to reflect essentially all incident RF energy at this frequency. By wrapping the water container in aluminium foil and leaving only two small apertures, one on each side of the container directly opposite each other, all propagation paths except the straight through-water path are blocked. The signal that reaches the Slave antenna has no choice but to have passed through the water column.

6.2 Implementation

The implementation procedure requires the following steps:

- Wrap the entire container in aluminium foil, ensuring complete coverage with no gaps along the seams. Overlap all edges by at least 10 mm and press firmly to maintain electrical continuity.
- Cut two apertures in the foil, each approximately 20 mm by 20 mm, positioned at diametrically opposite points at the same height on the container. These apertures define the measurement zone.
- Remove the pin header spacers and press the flat face of each ESP32-C3 Supermini board directly against the foil surface at each aperture, with the PCB antenna trace oriented vertically (parallel to the cup axis).
- Secure both boards firmly with multiple wraps of adhesive tape. The boards must not shift during measurement, as any movement changes the coupling geometry and introduces RSSI variation independent of turbidity.

In this configuration, the foil surrounding each aperture acts as a waveguide mouth, channelling the RF beam through the defined measurement window. The expected clean-water RSSI through the shielded configuration should be significantly lower (more negative) than the unshielded measurement, because the dominant low-loss air paths are removed. The turbidity-induced change will also follow the correct physics (RSSI increases with turbidity) but the change per NTU will be more consistent and reproducible because the measurement geometry is well-defined.

6.3 Expected Results and Limitations

With proper shielding in place, the measurement should behave as follows:

- Clean water: RSSI approximately -50 to -70 dBm depending on container diameter and board coupling.
- RSSI will still increase with turbidity due to particle displacement of water (the fundamental physics problem described in Section 5.1 is not resolved by shielding).
- The measurement will be reproducible and geometrically controlled, allowing a calibration curve to be established, though it will be an inverse curve (higher RSSI equals higher turbidity).

The critical limitation of this strategy is that the physical direction problem (RSSI increasing with turbidity) remains. The shielded measurement can be calibrated to infer turbidity from the RSSI increase, but the dynamic range at drinking water NTU levels (0-500 NTU) is expected to be only 1-3 dBm, which is comparable to the noise floor of the unfiltered RSSI measurement and only marginally above it with median filtering. This strategy is most useful for establishing a controlled baseline measurement and for gross turbidity detection rather than precision NTU quantification.

7. Alternative Strategy B: ESP32 Channel State Information (CSI) Phase Measurement

The second and more technically promising alternative strategy moves beyond amplitude-based RSSI measurement to phase-based measurement using Channel State Information (CSI) available from the ESP32's WiFi subsystem.

7.1 Why Phase Is More Sensitive Than Amplitude

The complex permittivity of water has two parts: the real part ϵ' governs the phase velocity of the wave, and the imaginary part ϵ'' governs attenuation. RSSI measures only the received amplitude, which reflects changes in ϵ'' . When turbidity changes, the primary dielectric change is in ϵ' (the particle displacement changes the real part of the mixture permittivity more significantly than the imaginary part at low volume fractions). RSSI is therefore measuring the less sensitive quantity.

Phase measurement accesses ϵ' directly. The phase of the received signal accumulates as the wave travels through the medium according to:

$$\phi = (2\pi d / \lambda_0) * \sqrt{\epsilon'}$$

Where d is the path length through the medium, λ_0 is the free-space wavelength, and ϵ' is the real part of the mixture permittivity. A change in ϵ' due to particle loading produces a proportional change in the accumulated phase, which is detectable from the CSI data.

7.2 What CSI Provides

When the ESP32 receives an 802.11n WiFi packet, the WiFi subsystem performs channel estimation across all OFDM subcarriers (52 data subcarriers in the 2.4 GHz band). This channel estimation produces a complex-valued channel frequency response for each subcarrier:

$$H[k] = A[k] * \exp(j * \phi[k])$$

Where $A[k]$ is the amplitude and $\phi[k]$ is the phase for subcarrier k . The RSSI-based approach discards all phase information and reduces all 52 subcarriers to a single average amplitude value. The CSI approach retains the full complex response, providing 52 amplitude values and 52 phase values per received packet, a total of 104 measurements per packet compared to 1 for RSSI.

This dramatically increases the information available about the propagation medium, and the phase component specifically gives access to the ϵ' changes that dominate at low turbidity concentrations.

7.3 Hardware Requirement

CSI extraction is not available on the ESP32-C3. The C3 uses a different WiFi hardware block that does not expose CSI to the application layer. CSI is available on the original ESP32 (Xtensa LX6 dual-core) and the ESP32-S3. The black 38-pin ESP32 DevBoard already in the project bill of materials supports CSI and can be used directly for this purpose without additional procurement.

Board	CSI Support	RSSI from rx_ctrl	Notes
ESP32-C3 Supermini	No	Yes	Used in current RSSI experiment; cannot be used for CSI
ESP32 DevKit (38-pin, black)	Yes	Yes	Already in project BOM; recommended for CSI implementation
ESP32-S3	Yes	Yes	Alternative option if S3 board is available

7.4 Implementation Architecture

The CSI implementation requires the following architectural changes relative to the current ESP-NOW RSSI system:

- Replace the Slave ESP32-C3 with the ESP32 DevKit for the receiver node.
- Enable CSI collection using the `esp_wifi_set_csi()` and `esp_wifi_set_csi_rx_cb()` API calls from the ESP-IDF WiFi driver.
- Configure the CSI callback to receive packets as IEEE 802.11 WiFi frames. The Master node transmits standard WiFi packets (not ESP-NOW) so that the 802.11n preamble is present and the receiver can perform OFDM channel estimation.
- In the CSI callback, extract the per-subcarrier phase values `phi[k]` from the `wifi_csi_info_t` structure.
- Apply a phase sanitisation step to remove the linear phase slope introduced by timing offset between transmitter and receiver. This is done by computing the differential phase between adjacent subcarriers, which cancels the common timing offset.
- Average the differential phase values across all subcarriers to produce a single phase measurement per packet.
- Apply the same median filtering and temperature compensation approach used in the RSSI system, then convert phase to TDS and turbidity via empirical calibration.

The core firmware addition to the receiver node is the CSI callback registration and processing:

```
esp_wifi_set_csi_rx_cb(csi_callback, NULL);
esp_wifi_set_csi(true);
```

```
void csi_callback(void *ctx, wifi_csi_info_t *info) {
    int8_t *csi_data = info->buf;
    // Extract imaginary and real parts per subcarrier
    // Format: [imag[0], real[0], imag[1], real[1], ...]
    for (int k = 0; k < info->len/2; k++) {
        float imag = (float)csi_data[2*k];
        float real = (float)csi_data[2*k + 1];
        float phi = atan2f(imag, real); // phase for subcarrier k
    }
}
```

```
    // store phi[k] for differential phase computation
  }
}
```

7.5 Expected Performance Improvement

The estimated performance improvement of CSI phase over RSSI amplitude is substantial. The phase change per NTU at 2.4 GHz through a 5 cm path is approximately 0.01 to 0.05 radians per 100 NTU, compared to the sub-1-dBm RSSI change for the same turbidity range. CSI phase resolution using averaging across 52 subcarriers is approximately 0.005 to 0.01 radians, giving an estimated detection threshold of 50 to 200 NTU. While this does not reach the 1 NTU precision of optical nephelometry, it is sufficient for a gross turbidity classification system (clean, moderately turbid, severely turbid) which is adequate for field water quality screening.

Metric	RSSI Method	CSI Phase Method
Measurements per packet	1 (amplitude only)	104 (52 amplitude + 52 phase)
Physical quantity measured	e'' (dielectric loss)	e' (dielectric constant)
Sensitivity to particle loading	Very low	Moderate
Estimated detection threshold	>5000 NTU	~50-200 NTU
Temperature sensitivity	High	Moderate
Hardware required	2x ESP32-C3 (current)	ESP32 DevKit + ESP32-C3 (1 swap)
Additional cost	None	None (DevKit already in BOM)
Implementation complexity	Low	High

7.6 Patent Significance of the CSI Method

The application of CSI phase measurement to water turbidity sensing within a cup-form-factor non-contact device is not represented in the prior art identified during the project's patentability review. The combination of: (a) a cup-shaped vessel with external-wall-mounted RF transceivers, (b) CSI-based differential phase extraction from WiFi OFDM subcarriers, and (c) empirical mapping of phase shift to turbidity in a multi-parameter water quality device, constitutes a novel and non-obvious method claim that strengthens the provisional patent specification beyond what the RSSI-only architecture could support.

8. Comparative Assessment of All Strategies

Strategy	Addresses Direction Problem	Addresses Magnitude Problem	Feasibility	Patent Value	Cost
RSSI (current)	No	No	Demonstrated — does not work as intended	Low	None
RSSI + Foil Shield	No	Partially	High — same hardware, foil only	Low-Medium	Negligible
RSSI + Longer Path (20-30 cm)	No	Partially	Medium — requires wider vessel	Low	None
LoRa at 868 MHz	Partially	Yes	Medium — new hardware required	Medium	~400 INR
CSI Phase (ESP32 DevKit)	Yes	Yes (partial)	Medium-High — firmware complexity	High	None (in BOM)

The RF shielding strategy is recommended as the immediate next step because it requires no new components, can be implemented within the current test session, and will produce controlled and reproducible measurements even if the dynamic range remains limited. The CSI strategy is recommended as the medium-term development path because it addresses the physics problem more fundamentally and produces the strongest patent claim.

9. Conclusion and Recommended Path Forward

The RF-RSSI turbidity sensing method, as implemented in the initial experiment, did not function as hypothesised. The primary failure was a fundamental physics constraint: at 2.4 GHz, suspended particles displace the water that is responsible for RF attenuation, causing RSSI to increase with turbidity rather than decrease. The secondary failure was insufficient signal magnitude at drinking water turbidity levels, given the 50 mm path length and 1 dBm resolution of the ESP32's RSSI reporting.

These findings do not invalidate the non-contact RF architecture as a patent claim. The method is novel, the apparatus is novel, and the experimental investigation has produced well-documented results. The patent specification should be framed honestly: the RF channel through the water wall responds to turbidity-induced changes in bulk permittivity, the direction and magnitude of that response at drinking water turbidity levels differs from the original hypothesis, and two specific improvements (shielding and CSI) are identified as the path to a functional implementation.

The recommended immediate actions are:

- Implement RF shielding using aluminium foil around the test bottle, press boards flush against foil apertures, and repeat the flour turbidity test to obtain geometrically controlled measurements.
- Record the RSSI (median) values at each turbidity level under shielded conditions and document the calibration curve showing RSSI increasing with NTU.
- Implement CSI collection on the ESP32 DevKit using the `esp_wifi_set_csi_rx_cb()` API and validate that differential phase values change with turbidity in a controlled measurement cell.
- Update the provisional patent specification to include both the amplitude-based and phase-based embodiments as alternative claims within the same non-contact RF turbidity sensing architecture.

Key Finding Summary

The RSSI method does not work as originally hypothesised at drinking water turbidity levels. The physics is confirmed and documented. The CSI phase method addresses the fundamental limitations and is the recommended path forward. The RF architecture retains strong patent value as a novel approach to non-contact turbidity sensing in a cup-form-factor device.

End of Report